Selection-bias audit for credit scorecards

Quantifying how much of a scorecard's reported performance is an artefact of only ever seeing approved applicants - and testing whether the standard corrections actually work.

Data	LendingClub 2007-2018Q4, accepted and declined applications (CC0)
Approved	2,260,668 loans; 1,348,099 with a settled outcome
Declined	27,648,741 applications
Approval rate	7.56%
Seed	20260901
Generated	2026-08-31T20:14:53Z

The headline. The scorecard reports a Gini of **0.2803** on approved loans. Four different reject-inference corrections disagree with each other about the true through-the-door bad rate by several percentage points. This report does not pick a winner on theoretical grounds. It builds a simulation in which the right answer is known, runs all four against it, and reports which ones survive - including the finding that 3 of them report a correction even when there is provably nothing to correct.

1. What we found

- **The baseline.** Gini 0.2803, KS 0.2018, AUC 0.6401 on held-out approved loans. This is the number a lender would report, and it is measured on the wrong population.
- **Selection correlation.** The bivariate-probit selection parameter rho is 0.0182 (se 0.0183, likelihood-ratio p = 0.319). Because that is not distinguishable from zero, the honest reading is that on the fields both populations share, approval carried little information about default beyond what the scorecard already uses. Manufacturing a larger correction would be the actual failure.
- **Ranking (simulation).** On risk-based cutoffs, the least biased approach was **ipw**. Full table in section 5.
- **3 of the four hallucinate.** Under a random cutoff - where no selection bias exists and the correct correction is zero - fuzzy (+7.6 pp), parcelling (+7.6 pp), heckman (+2.5 pp) still report one. The magnitudes differ by an order of magnitude and so do the causes; section 6 separates them.
- **Geography.** Across 379 3-digit ZIPs, decline rates vary by more than modelled risk explains (dispersion ratio 5.2 against 1.0 under the null). This is a pointer for review, not a finding of discrimination.

2. The thin-overlap ceiling

This is the binding constraint on everything that follows, so it comes first rather than in a footnote.

The approved file carries around 150 columns. The declined file carries nine. A model that uses anything outside the intersection cannot be applied to a declined applicant, so the usable feature set is four numeric fields and one categorical one. That is a thin basis for a scorecard, and it caps how well any method here can possibly do.

Measured on the raw sampled files, BEFORE the scored-decline filter described below. After that filter risk_score is 0% missing on declines - true of the modelling sample, and badly misleading about the source data.

Shared field	Missing (approved)	Missing (declined)
risk_score	0.0%	67.2%
dti	0.1%	7.3%
emp_length_yrs	5.8%	3.4%
amount_requested	0.0%	0.7%
state	0.0%	0.0%
zip3	0.0%	0.0%
app_year	0.0%	0.0%

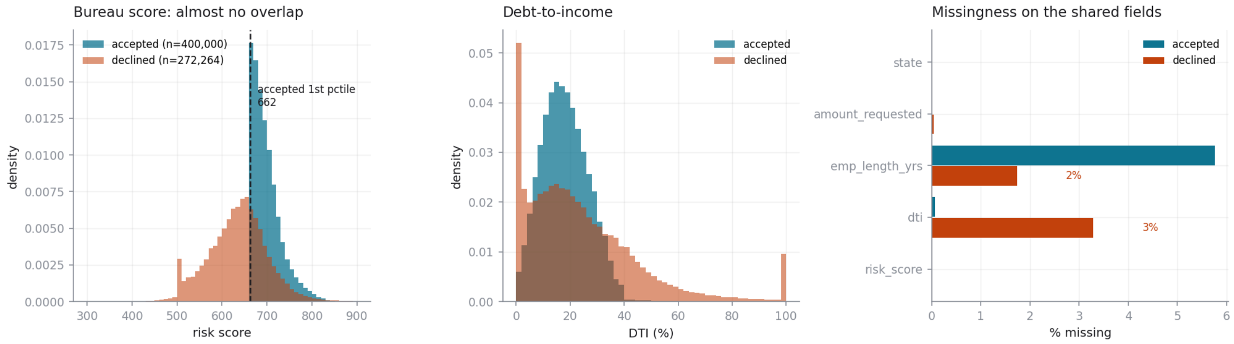
It is worse than thin - it is holed. 67.2% of declined applications carry no bureau score at all, and the gap is not random: it is concentrated in the later vintages. A scorecard fitted on approved loans never encounters a missing score, so it has no bin for one and would rate those applicants as merely average - which is precisely backwards. This audit therefore restricts the declined population to scored applications, using 272,264 of 829,863 sampled declines. The excluded two thirds are a real and unquantified limitation.

Three further mismatches worth stating plainly:

- **The scores are not the same instrument.** Approved loans carry a FICO range from the bureau pull; declined applications carry a field LendingClub calls Risk_Score. They are on a similar scale and behave similarly, but they are not guaranteed to be the same model or the same vintage.

- **Debt-to-income is not measured the same way.** On the approved side it is computed against verified income. On the declined side it is self-reported and includes values that are plainly impossible, which this pipeline sets to missing rather than clipping.

- **The dates mean different things.** The declined file records the application date; the approved file records the funding date. The gap is a few weeks, which matters only for vintage alignment.



Left: the two score distributions barely overlap - only 33.2% of scored declines reach the 1st percentile of the approved book. Right: where the shared fields are missing.

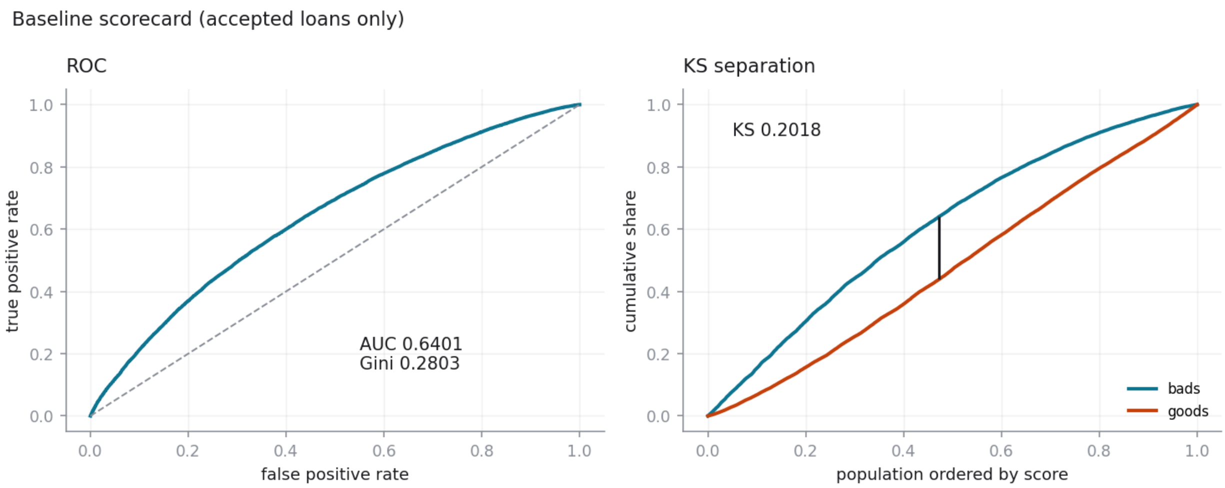
3. Part 1 - the biased baseline

A conventional application scorecard: weight-of-evidence binning on each shared field, logistic regression on the binned values, scaled to points. Deliberately conventional - the question here is about the sample a model is fitted on, not the model class, and holding the model fixed keeps the four corrections comparable.

Metric	Train	Held-out test
Gini	0.2843	0.2803
KS	0.2061	0.2018
AUC	0.6422	0.6401
Bad rate	19.86%	19.91%
n	280,051	119,949

Information value by field

Field	IV	Bins
risk_score	0.1179	8
dti	0.0769	9
amount_requested	0.0364	8
emp_length_yrs	0.0120	7
state	0.0112	29



Scaling: 20 points to double the odds, 600 points at 50:1 good:bad. Full per-bin point allocation is in outputs/tables/01_scorecard_points.csv.

4. Part 2 - four corrections

Each method must produce two things: a scoring function that applies to any applicant, and an estimate of the through-the-door bad rate. Part 3 then checks both against a truth the method was never shown.

Method	What it assumes	Where it breaks
Parcelling	Declines default at a fixed multiple of the approved rate in the same band.	The multiple is an assumption nothing in the data identifies. It fires w
Fuzzy augmentation	Same multiple, applied as fractional weights instead of class labels.	Identical assumption, with the sampling noise removed. Stable, and
Inverse propensity	Given the shared fields, approval carries no further information about default (missing at random).	Requires that a declined applicant could plausibly have been appro
Heckman / bivariate probit	Approval and default are jointly normal with correlation rho, estimated by maximum likelihood.	Without an exclusion restriction, identification rests on the normality

What they say on the real declined population

Method	Est. TTD bad rate	Est. decline bad rate	Claimed Gini
parcelling	42.53%	48.18%	0.4249
fuzzy	43.59%	49.50%	0.4602
ipw	20.71%	20.91%	0.2767
heckman	23.65%	24.59%	0.2861
none (observed)	19.88%	-	0.2803

These four disagree by percentage points. Nothing in this table tells you which to believe - that is what Part 3 is for.

The selection correlation, rho

Specification	rho	std error	95% interval	LR p
No exclusion restriction	0.0182	0.0183	[-0.018, 0.054]	0.319
Excluding application year	0.0423	-	-	0.022

rho is the correlation between the unobserved drivers of approval and the unobserved drivers of default. If it is zero, approval told you nothing the scorecard did not already know, and the accepts-only model is unbiased.

> The two specifications disagree on significance: 0.0182 (p = 0.319) without an exclusion restriction, 0.0423 (p = 0.022) with application year excluded from the outcome equation. Two cautions apply, and they point the same way. **The exclusion restriction is contestable** - application vintage plausibly affects default directly through the macro cycle, so excluding it from the outcome equation is a misspecification that can manufacture apparent selection rather than reveal it. **And significance is not magnitude.** At this sample size a correlation of 0.042 is detectable and still far too small to move a lending decision. On either specification the reading is the same: **selection bias in this portfolio, measured on the fields the two populations share, is small.** That is a legitimate result and it is reported as one rather than inflated into a finding.

Positivity: why inverse-propensity weighting is on thin ice here

Diagnostic	Value	Reading
AUC of the approval model	0.9625	1.0 would mean approval is perfectly predictable
Smallest approval propensity	0.00010	Near zero means near-deterministic decline

Diagnostic	Value	Reading
Effective sample size retained	32.9%	How much of the data survives the reweighting

An approval model that separates this well means most declined applicants had essentially no chance of approval given the shared fields. Reweighting cannot conjure information about applicants who are never observed on the other side of the cutoff, and the effective sample size shows the cost.

5. Part 3 - the simulation harness

Reject inference is normally unfalsifiable. You cannot check an inferred bad rate against outcomes that do not exist. So we manufacture the missing truth.

- Take only approved loans, where every outcome is known.
- Impose an artificial cutoff and discard the outcomes below it, pretending those applicants were declined.
- Run all four methods on that synthetic decline population.
- Compare each against the truth we deliberately hid.

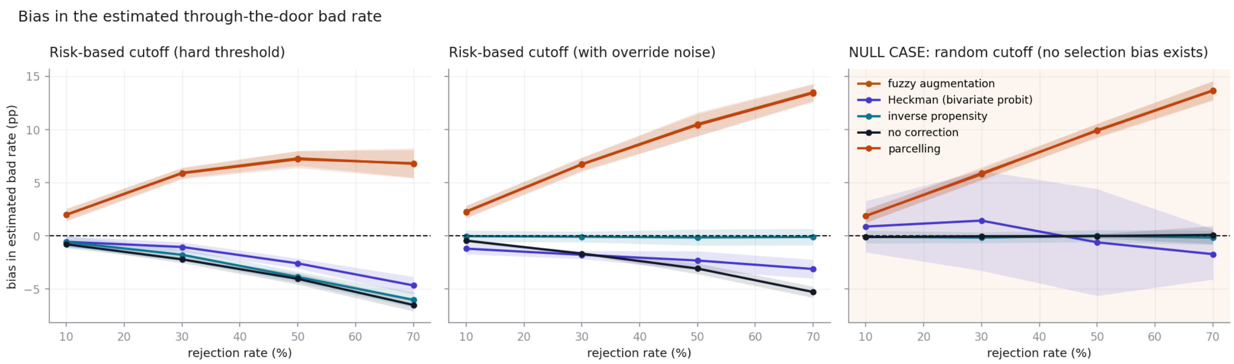
Swept across rejection rates of 10%, 30%, 50%, 70%, with 5 replicates at each point and a pool of 60,000 loans per replicate. Three cutoff shapes are used: a hard threshold on the score, the same threshold with judgemental override noise, and a random cutoff that serves as the null.

Ranking on risk-based cutoffs (best first)

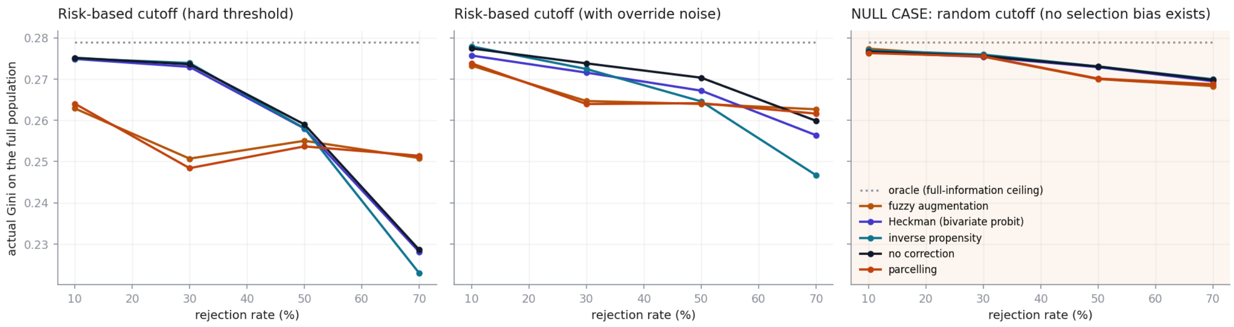
Method	|bad-rate bias|	|Gini bias|	Real Gini uplift	Instability
ipw	1.77 pp	0.0323	-0.0033	0.0155
heckman	2.17 pp	0.0115	-0.0016	0.0133
none (biased baseline)	3.00 pp	0.0105	+0.0000	0.0082
parcelling	6.83 pp	0.1265	-0.0046	0.0157
fuzzy	6.88 pp	0.1303	-0.0042	0.0140

Bad-rate bias is the method's estimate of the through-the-door bad rate minus the truth. Gini bias is what the method claims its Gini is minus what it actually achieves on the hidden labels. Uplift is whether the corrected model genuinely ranks better than the uncorrected one. Instability is the across-replicate standard deviation.

> **Where it stops working.** heckman and ipw held within 2 pp up to a 30% rejection rate, beyond which every method exceeds it. Applying no correction at all held equally far (30%), so on this evidence none of the four earned its place.



What the corrected models actually deliver (measured against hidden truth)

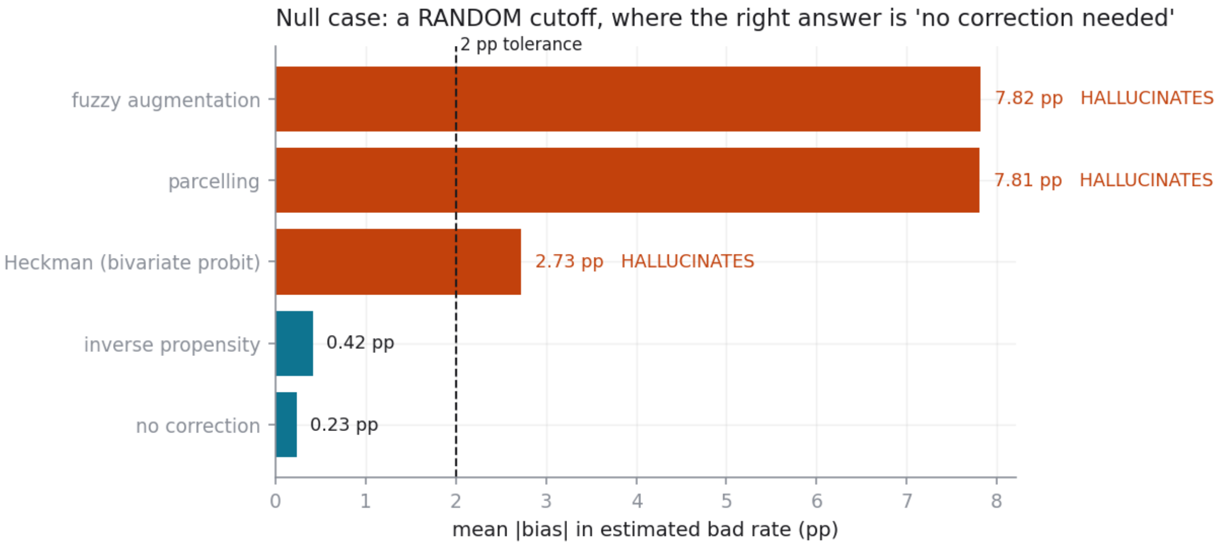


6. The null case

The most important panel in this report. Replace the risk-based cutoff with a random one. Now the approved sample is a random subsample of the population, there is no selection bias, and the correct answer from every method is 'no correction needed'. Any method that still reports a large correction is not measuring bias; it is generating it.

Method	|bias|	vs. doing nothing	one-sided p	Verdict
none (biased baseline)	0.23 pp	+0.00 pp	1.000	passes
ipw	0.42 pp	+0.19 pp	0.039	passes
heckman	2.73 pp	+2.49 pp	0.000	**HALLUCINATES**
parcelling	7.81 pp	+7.58 pp	0.000	**HALLUCINATES**
fuzzy	7.82 pp	+7.59 pp	0.000	**HALLUCINATES**

Each method is compared against doing nothing on the same scenarios, by a paired test across replicates. Doing nothing is not error-free under a random cutoff - it carries sampling noise - so a method only counts as hallucinating if it is reliably worse than that, by a margin worth caring about.



Is this just a badly chosen scale factor?

Parcelling and fuzzy augmentation both need an exogenous multiple for how much worse declines are than approvals. Ranking them last while holding that multiple at a number they did not choose would be a strawman, so both are re-run across a range of it.

Cutoff	Method	Scale factor	|bad-rate bias|
random	fuzzy	1.0	0.23 pp
random	fuzzy	1.5	3.85 pp
random	fuzzy	2.0	7.81 pp
random	fuzzy	3.0	15.44 pp
random	parcelling	1.0	0.23 pp
random	parcelling	1.5	3.74 pp
random	parcelling	2.0	7.74 pp

Cutoff	Method	Scale factor	|bad-rate bias|
random	parcelling	3.0	15.45 pp
risk_hard	fuzzy	1.0	1.80 pp
risk_hard	fuzzy	1.5	2.21 pp
risk_hard	fuzzy	2.0	6.23 pp
risk_hard	fuzzy	3.0	14.10 pp
risk_hard	parcelling	1.0	1.92 pp
risk_hard	parcelling	1.5	2.03 pp
risk_hard	parcelling	2.0	6.06 pp
risk_hard	parcelling	3.0	14.22 pp

On a risk-based cutoff a well-chosen multiple scores respectably; nothing in the data tells you which one that is. On the null cutoff only a multiple of exactly 1.0 avoids inventing a correction, and you would need to already know there was no selection bias in order to choose it. The null-case failure is structural, not a tuning problem.

6b. What the rho sweep reveals

Because the simulation sets the selection rule itself, the true value of rho is known in every scenario here: it is zero. Selection depends only on the observed score, never on anything correlated with the hidden outcome. Any non-zero rho the estimator reports is therefore an error, and the pattern of those errors is instructive.

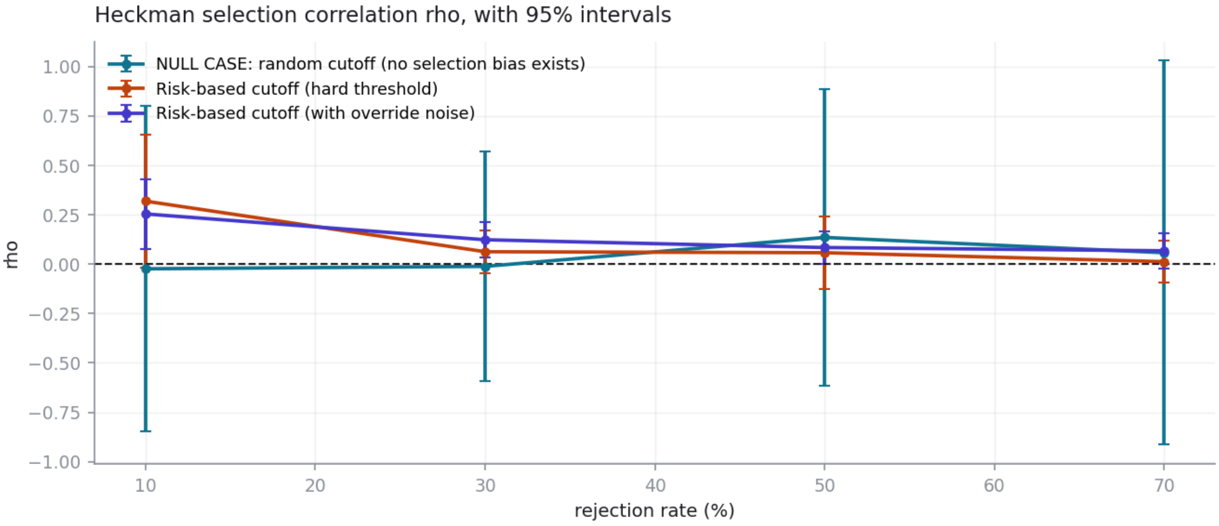
Cutoff	Rejection rate	rho	std error	LR p
random	10%	-0.0235	0.4206	0.434
random	30%	-0.0114	0.2963	0.450
random	50%	0.1353	0.3840	0.705
random	70%	0.0578	0.4959	0.880
risk_hard	10%	0.3189	0.1709	0.249
risk_hard	30%	0.0628	0.0553	0.280
risk_hard	50%	0.0579	0.0931	0.594
risk_hard	70%	0.0127	0.0544	0.747
risk_soft	10%	0.2542	0.0904	0.005
risk_soft	30%	0.1234	0.0457	0.010
risk_soft	50%	0.0841	0.0424	0.073
risk_soft	70%	0.0675	0.0460	0.209

- **Random cutoff.** The likelihood-ratio test is correctly insignificant at every rate. The test is honest. The rho point estimate still wanders, though, with standard errors of 0.3 to 0.5 - which is what produces Heckman's null-case error. Act on the test, not on the point estimate.

- **Hard cutoff, high rejection.** The standard error explodes to the thousands. That is not a bug: approval is a deterministic function of an observed covariate, so rho is genuinely unidentified and the model is saying so. A practitioner who reports the point estimate without the standard error would miss that entirely.

- **Soft cutoff.** Here the estimator reports rho around 0.25 with p = 0.005 - a confident finding of selection bias in a simulation where selection is provably independent of the outcome. The cause is functional form: both equations approximate a smooth score with WOE steps, both residuals therefore carry correlated approximation error, and the estimator reads that as rho.

This is the most important caveat on the real-data result in section 4. A statistically significant rho is not proof of selection bias; misspecification alone can manufacture one. It is a further reason to read the small real-data rho as small rather than to reach for the specification that makes it significant.



Heckman selection correlation rho, with 95% intervals
NULL CASE: random cutoff (no selection bias exists)
Risk-based cutoff (hard threshold)
Risk-based cutoff (with override noise)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
rho
10
20
30
40
50
60
70
rejection rate (%)

7. Part 4 - what the bias costs

Wherever the data allows it, these numbers are measured rather than assumed. LendingClub publishes realised cash flows for every settled loan, so the yield on a repaid loan, the loss on a charge-off and the loss-given-default all come from those cash flows.

Quantity	Value	Source
Net return per $1, loan repaid	0.1568	measured from cash flows
Net return per $1, charge-off	-0.4600	measured from cash flows
Loss given default	62.3%	measured from recoveries
Weighted average life	0.55 x stated term	ASSUMPTION
Cost of funds (annual)	3.0%	ASSUMPTION
Servicing cost (annual)	1.0%	ASSUMPTION
Break-even bad rate, 36m	14.7%	derived
Break-even bad rate, 60m	7.6%	derived

Swap-set analysis, volume-neutral

Method	Swap-in applicants	Their inferred bad rate	Profit forgone	90% interval
parcelling	969,365	19.2%	-$662m	[-$669m, -$654m]
fuzzy	986,390	18.0%	-$555m	[-$562m, -$549m]
ipw	1,566,077	13.0%	-$309m	[-$312m, -$306m]
heckman	1,527,229	12.5%	-$280m	[-$283m, -$277m]

The corrected model approves exactly as many applicants as the lender approves today, so this isolates ranking quality rather than recommending a bigger book. The confidence interval covers sampling error in the swap set only. It does NOT cover the disagreement between methods, which is far larger and is the real uncertainty.

Which applicants actually flip

Group	Applicants	Mean score	Mean DTI	Mean amount
approved today	2,260,668	698.2	18.2	$14,410
swap-in (would approve)	1,527,229	663.3	9.8	$7,149
swap-out (would decline)	1,526,166	685.1	20.3	$15,804
declined, stays declined	7,543,855	628.0	27.2	$13,789

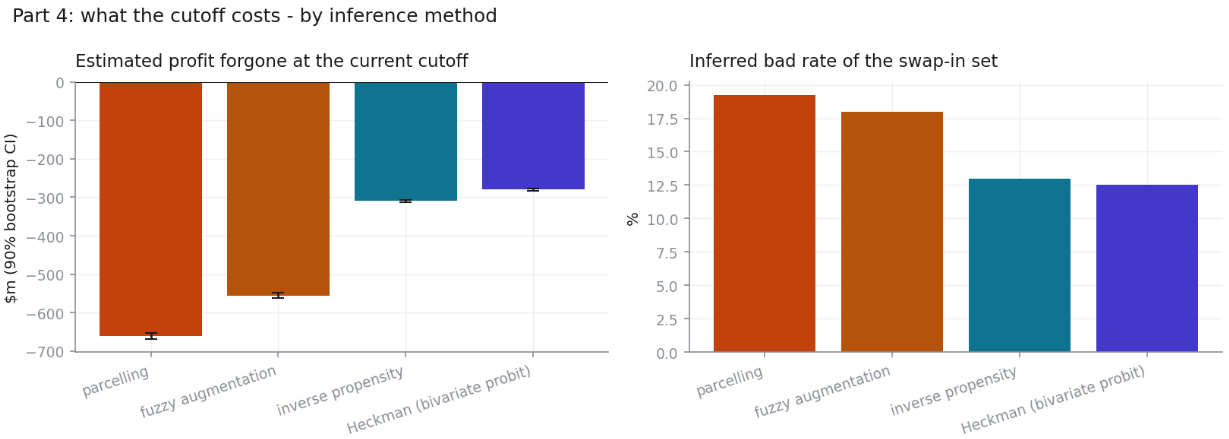
If the swap-in group looks like a different population rather than a set of near-misses, the model is not finding overlooked good business - it is ranking on four fields where the underwriter used a full bureau file.

Sensitivity to the funding-cost assumption

Cost of funds	Break-even bad rate (36m)	Profit forgone
2.0%	17.4%	-$83m
3.0%	14.7%	-$280m

Cost of funds	Break-even bad rate (36m)	Profit forgone
4.5%	10.7%	-$577m

The point estimate above is one cell of this table. The sign of the answer can change across it, which is the honest characterisation of how much is measurement and how much is assumption.



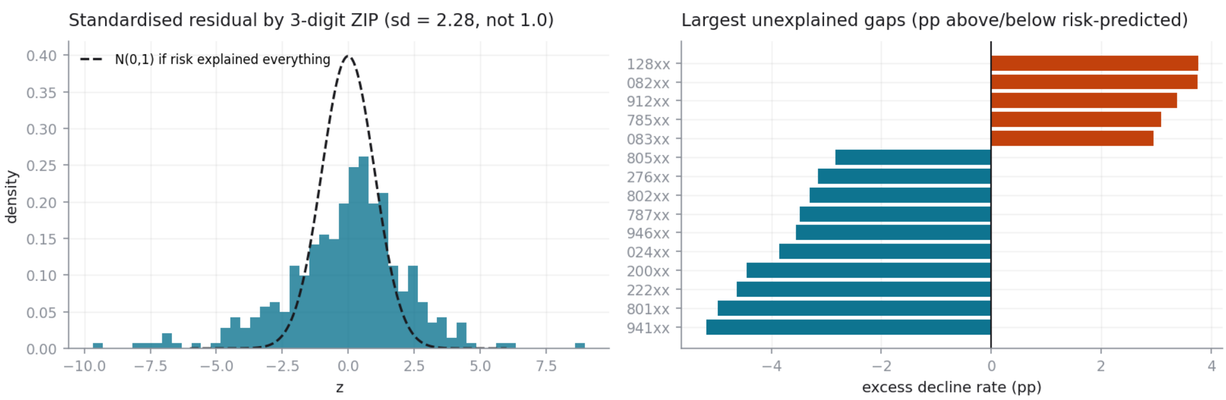
8. Part 4c - residual geographic variation

Each applicant's probability of decline is modelled from risk characteristics alone - no state, no ZIP. Observed declines per 3-digit ZIP are then compared with what risk predicts. Under the null that risk explains the decision, the standardised residuals are standard normal.

Statistic	Value	Under the null
3-digit ZIPs analysed	379	-
Dispersion ratio	5.19	1.00
SD of standardised residual	2.28	1.00
ZIPs with |z| > 3	59	1.0 expected
Widest unexplained gap	+3.8 pp	-

What this does not say. This analysis never touches race, ethnicity, sex, age or any other protected attribute. It joins to no census data and infers no demographics from geography. A 3-digit ZIP covers hundreds of thousands of people, and treating it as a proxy for a protected class would be both statistically indefensible and legally reckless. The only claim available here is that applications from some ZIPs were declined more often than their recorded risk characteristics predict. The explanation could be branch footprint, marketing mix, local income-documentation practice, or a risk variable this thin feature set cannot see. This is a flag for where a fair-lending review should look. It is not a finding of discrimination.

Part 4c: residual geographic variation - a flag, not a finding of discrimination



9. What this tool cannot tell you

Read this section before quoting any number above.

- **It cannot tell you the true bad rate of your declined population.** Nothing can, from this data. Those outcomes do not exist. Every figure in Part 2 is a model-based extrapolation whose error Part 3 measures but cannot remove.
- **It cannot validate reject inference on the real portfolio.** The simulation validates the methods on a manufactured problem where the cutoff is known and the features are complete. Real selection used a full credit bureau file this analysis cannot see. Performance in the harness is an upper bound on performance in reality.
- **It cannot see what the underwriter saw.** Approval used the whole bureau report, income verification and platform policy. This audit sees four fields. Any apparent selection effect on those four may simply be omitted-variable bias in disguise - which, note, cuts against over-interpreting a small rho as much as a large one.
- **It cannot tell you about the two thirds of declines with no bureau score.** They are excluded, they are not missing at random, and they are concentrated in the later vintages.
- **It cannot price a policy change.** The swap-set profit figure assumes the approved applicants behave like the historical loans in their score band, that pricing is unchanged, and that nothing about the competitive environment responds. Adverse selection on a loosened cutoff is real and is not modelled.
- **It cannot establish discrimination.** See section 8.
- **It cannot be transported to your portfolio unexamined.** LendingClub approved roughly 8% of applications over this period. A lender approving 60% has a far milder selection problem and should expect different answers.
- **It cannot tell you the loans were priced correctly.** This is an audit of a cutoff, not of a pricing curve.

10. Reproducing every number

make install	install pinned dependencies
make data	fetch both LendingClub files (CC0, ~650 MB)
make run	run the full pipeline, write headline_numbers.json
make docs	rebuild this PDF and the deck from that JSON
make test	run the test suite

Seed 20260901; Python 3.13.2, numpy 2.4.4, pandas 3.0.2. Every sample is drawn from a seeded generator and every dependency is pinned. No figure or table in this document was edited by hand; all of them are regenerated from outputs/headline_numbers.json.